

Major League Baseball Database

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Professional Sports and League Management: Major League Baseball

In the data-driven world of Major League Baseball (MLB), league administrators, teams, players, coaches, and trainers all depend on accurate and timely information to be the best they can be. Data is central to almost every aspect of the sport, from tracking long-term player development to handling team rosters and player contracts to analyzing in game performance. In order to solve key data management challenges in the MLB, this project focuses on developing a relational database to address said challenges.

Data Management Problems

Professional baseball's current data management systems tend to be inconsistent, which can lead to:

- **Inconsistent Player Statistics:** Analysts and coaches can get confused by conflicting or inaccurate player stats.
- **Inaccurate Injury Tracking:** Athlete injury records and recovery time could be inaccurately recorded due to the lack of a central tracking framework.
- **Contract and Roster Issues:** Inaccurate information related to contracts can result in delays in contract signing, salary cap violations, or payroll discrepancies.
- **Scheduling Conflicts:** Scheduling without the use of accurate venue and team availability data can cause scheduling issues.
- **Performance Analysis Delays:** Coaches and front offices may not be able to quickly access player or game data when making decisions.

Motivation

This project aims to implement a relational database that addresses the above issues by using core database principles:

- **Data Integrity:** Enforcing primary and foreign key constraints guarantees consistency across tables, avoiding data inconsistencies (for example, a player playing for two teams at the same time).

- **Reduced Data Redundancy:** Storing player and team information in normalized relations makes sure that repetition and errors throughout the system are prevented.
- **Faster Access and Querying:** Structured data allows for faster queries, like filtering for all hitters that have/had a batting average higher than .300 in a particular season.
- **Scalability:** The database will grow as the league itself grows. It can handle expansion teams, new more detailed statistics, or changes to contracts.

Potential Benefits

Implementing this database could have the following real-world benefits:

- **Efficiency in Operations:** Managers can easily update rosters, coaches can track player availability, and staff can manage contracts without redundant work.
- **Better Decision-Making:** Coaches and the front office can make educated decisions in real-time with access to live player statistics and data (injury status, etc).
- **Accurate Reporting:** Media outlets and league officials can query the same database for accurate and reliable statistics.
- **Data-Driven Strategy:** Front offices can use historical data and trends to decide on player acquisitions, game strategies, and training plans.

Potential Users

- **Team Managers** – They can keep track of player movement, contract status, and roster compliance.
- **Coaches & Analysts** – They can use game and player data to optimize lineups and strategies.

- **League Officials** – They can keep an eye on schedules, team compliance, and rule enforcement.
- **Medical Staff** – They can record injuries and track player recovery.
- **Media Personnel** – They can make reports using reliable and current data.

Business Rules

1. Player-Team Assignment

MLB teams each manage rosters of players who sign contracts to play for them.

- A player can only be assigned to one team at a time.
- A team can have many players.
- Player records: PlayerID, FirstName, LastName, Position, BirthDate, Nationality, Bats, Throws.

2. Contracts

Players sign contracts that say their salary and for how long.

- A contract is connected to one player and one team.
- A player can have many contracts over time but only one current contract.
- Attributes: ContractID, StartDate, EndDate, Salary.

3. Games and Seasons

- Each game is associated with one season.
- Each season has many games (typically 162 games, if there are no strikes / lockouts).
- Game data: GameID, Date, HomeTeamID, AwayTeamID, Stadium, FinalScore.
- Season data: SeasonID, Year, LeagueChampion, MVP.

4. Player Performance (Stats)

Player performance is tracked per game, with detailed statistics.

- PlayerStats is an associative entity between Player and Game.
- Each record includes: AtBats, Hits, Runs, HomeRuns, RBIs, Strikeouts, Walks, Errors, InningsPitched.
- There has to be a PlayerStats record for each player in each game, even if all fields are zero.

5. Person Roles (Supertype / Subtype)

All individuals in the system are a Person.

- a. A Person can be a Player, Coach, Manager, or none.
- b. Attributes: PersonID, Name, DateOfBirth, Nationality.

6. Team Structure

- a. A team can have many games, either at home or away.
- b. There must be two teams in every game.
- c. A team is defined by: TeamID, TeamName, City, Division, League.

User Requirements

Access of reliable, timely data/stats is extremely essential for MLB officials, team staff, and analysts. The system must be able to support the following:

1. Team Managers

- Track player contracts, roster status, and salary agreements.
- View current and past contracts for each player.

2. Coaches and Analysis

- View detailed player stats by career, season, or game.
- Compare player statistics by position or team.

3. League Officials

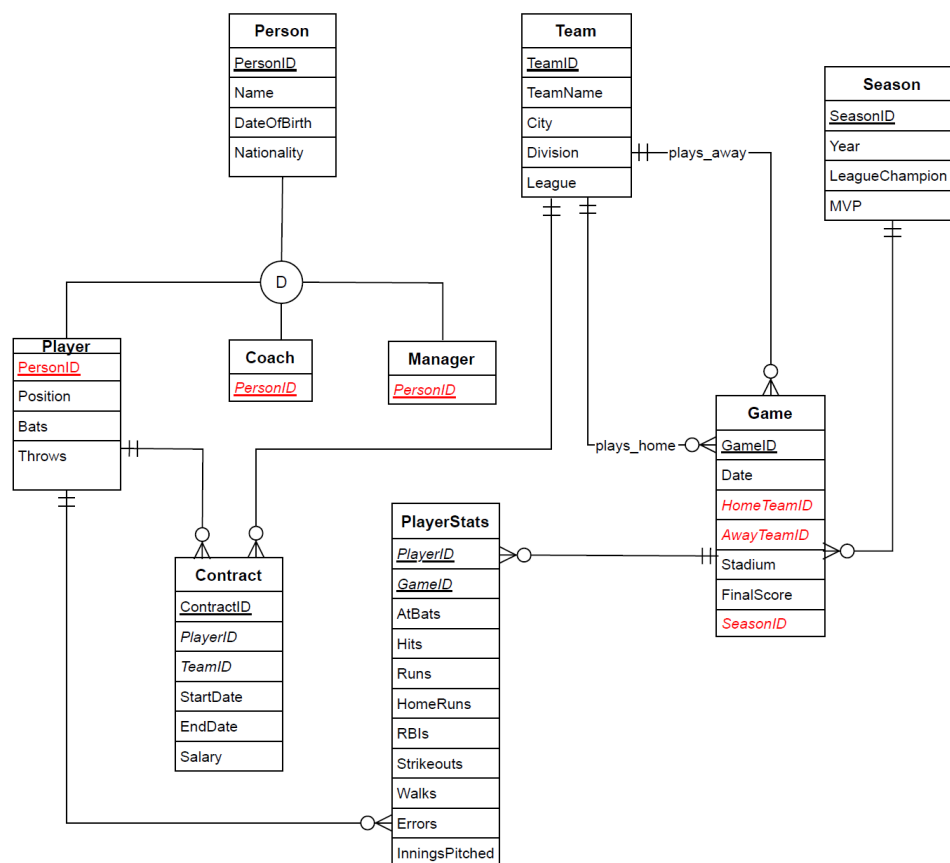
- Schedule games and make sure of roster compliance.
- Track league winners, Most Valuable Player (MVP), and season stats.
- Conduct queries on media sharing and statistical reporting.

4. System Actions Include:

- Viewing and changing player-team assignments.
- Recording game results and individual performance.
- Creating and updating contracts.
- Determining top performers players using aggregate queries

EERD Diagram

Format	Meaning
<u>Underlined</u>	Primary Key (PK)
<i>Italics + Red</i>	Foreign Key (FK)



Entities

- **Person**: the general entity for all members involved in the organization (players, coaches, managers). Contains personal information.
 - **Player**: A subtype of Person. Contains baseball-specific information like position and which side they hit and throw..

- Coach: Another subtype of Person. Represents the coaching staff.
- Manager: A third subtype of Person. Includes people who manage the team and make decisions for the team.
- Team: Contains data about the MLB teams, including city, league, and division.
- Contract: Tracks agreements between players and teams. Includes dates and salaries.
- Season: Contains statistics for a particular season, including MVP and league champion.
- Game: Represents an MLB game and includes home and away team, scores, and what season the game is taking place in.
- PlayerStats: An associative entity that contains the statistics of a player.

Relations

Person

<u>PersonID</u>	Name	DateOfBirth	Nationality
-----------------	------	-------------	-------------

Player

<i><u>PersonID</u></i>	Position	Bats	Throws
------------------------	----------	------	--------

Coach

<i><u>PersonID</u></i>

Manager

<i><u>PersonID</u></i>

Team

<u>TeamID</u>	TeamName	City	Division	League
---------------	----------	------	----------	--------

Contract

<u>ContractID</u>	<i>PersonID</i>	<i>TeamID</i>	StartDate	EndDate	Salary
-------------------	-----------------	---------------	-----------	---------	--------

Season

<u>SeasonID</u>	Year	LeagueChampion	MVP
-----------------	------	----------------	-----

Game

<u>GameID</u>	Date	<i>HomeTeamID</i>	<i>AwayTeamID</i>	Stadium	FinalScore	<i>SeasonID</i>
---------------	------	-------------------	-------------------	---------	------------	-----------------

PlayerStats

<i><u>PlayerID</u></i>	<i><u>GameID</u></i>	AtBats	Runs	HomeRun	RBI	Strikeout	Walks	Errors	Innings Pitched
------------------------	----------------------	--------	------	---------	-----	-----------	-------	--------	--------------------

SQL Tables

```
CREATE TABLE Person (
```

```
    PersonID INT PRIMARY KEY,
```

```
    Name VARCHAR(100),
```

```
    DateOfBirth DATE,
```

```
    Nationality VARCHAR(50)
```

```
);
```

```
CREATE TABLE Player (
```

```
    PersonID INT PRIMARY KEY,
```

```
    Position VARCHAR(50),
```

```
    Bats VARCHAR(10),
```

```
    Throws VARCHAR(10),
```

```
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID)
```

```
);
```

```
CREATE TABLE Coach (  
    PersonID INT PRIMARY KEY,  
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID)  
);
```

```
CREATE TABLE Manager (  
    PersonID INT PRIMARY KEY,  
    FOREIGN KEY (PersonID) REFERENCES Person(PersonID)  
);
```

```
CREATE TABLE Team (  
    TeamID INT PRIMARY KEY,  
    TeamName VARCHAR(100),  
    City VARCHAR(100),  
    Division VARCHAR(50),  
    League VARCHAR(50)  
);
```

```
CREATE TABLE Contract (  
    ContractID INT PRIMARY KEY,  
    PlayerID INT,  
    TeamID INT,  
    StartDate DATE,  
    EndDate DATE,  
    Salary DECIMAL(10,2),  
    FOREIGN KEY (PlayerID) REFERENCES Player(PersonID),  
    FOREIGN KEY (TeamID) REFERENCES Team(TeamID)  
);
```

```
CREATE TABLE Season (  
    SeasonID INT PRIMARY KEY,  
    Year INT,  
    LeagueChampion VARCHAR(100),  
    MVP VARCHAR(100)  
);
```

```
CREATE TABLE Game (  
    GameID INT PRIMARY KEY,
```

```
Date DATE,  
  
HomeTeamID INT,  
  
AwayTeamID INT,  
  
Stadium VARCHAR(100),  
  
FinalScore VARCHAR(10),  
  
SeasonID INT,  
  
FOREIGN KEY (HomeTeamID) REFERENCES Team(TeamID),  
  
FOREIGN KEY (AwayTeamID) REFERENCES Team(TeamID),  
  
FOREIGN KEY (SeasonID) REFERENCES Season(SeasonID)  
  
);
```

```
CREATE TABLE PlayerStats (  
  
    PlayerID INT,  
  
    GameID INT,  
  
    AtBats INT,  
  
    Hits INT,  
  
    Runs INT,  
  
    HomeRuns INT,  
  
    RBIs INT,  
  
    Strikeouts INT,
```

```
Walks INT,  
  
Errors INT,  
  
InningsPitched DECIMAL(5,2),  
  
PRIMARY KEY (PlayerID, GameID),  
  
FOREIGN KEY (PlayerID) REFERENCES Player(PersonID),  
  
FOREIGN KEY (GameID) REFERENCES Game(GameID)  
  
);
```

Data for SQL Tables

-- Teams

```
INSERT INTO Team (TeamID, TeamName, City, Division, League)  
VALUES (1, 'Toronto Blue Jays', 'Toronto', 'AL East',  
        'American');  
  
INSERT INTO Team VALUES (2, 'New York Yankees', 'New York',  
        'AL East', 'American');
```

-- Seasons

```
INSERT INTO Season (SeasonID, Year, LeagueChampion, MVP)  
VALUES (2025, 2025, 'Houston Astros', 'Shohei Ohtani');
```

-- Games

```
INSERT INTO Game (GameID, Date, HomeTeamID, AwayTeamID,  
Stadium, FinalScore, SeasonID) VALUES (1001, '2024-03-28',  
1, 2, 'Rogers Centre', '5-3', 2024);
```

```
INSERT INTO Game VALUES (1002, '2024-03-30', 2, 1, 'Yankee  
Stadium', '4-2', 2024);
```

-- Season

```
INSERT INTO Season (SeasonID, Year, LeagueChampion, MVP)  
VALUES (2024, 2024, 'Los Angeles Dodgers', 'Shohei Ohtani');
```

```
INSERT INTO Season VALUES (2023, 2023, 'Texas Rangers',  
'Shohei Ohtani');
```

-- Managers and Coaches (Blue Jays)

```
INSERT INTO Person VALUES (1001, 'John Schneider',  
'1980-02-14', 'USA');
```

```
INSERT INTO Manager VALUES (1001);
```

```
INSERT INTO Person (PersonID, Name, DateOfBirth,  
Nationality) VALUES (1002, 'Don Mattingly', '1961-04-20',  
'USA');
```

```
INSERT INTO Coach (PersonID) VALUES (1002);
```

```
INSERT INTO Person VALUES (1003, 'Guillermo Martínez',  
'1984-11-05', 'USA');
```

```
INSERT INTO Coach VALUES (1003);
```

```
INSERT INTO Person VALUES (1004, 'Pete Walker',  
'1969-04-08', 'USA');
```

```
INSERT INTO Coach VALUES (1004);
```

```
INSERT INTO Person VALUES (1005, 'Mark Budzinski',  
'1973-08-20', 'USA');
```

```
INSERT INTO Coach VALUES (1005);
```

```
INSERT INTO Person VALUES (1006, 'Luis Rivera',  
'1964-01-03', 'Puerto Rico');
```

```
INSERT INTO Coach VALUES (1006);
```

```

-- Managers and Coaches(Yankees)

INSERT INTO Person VALUES (2001, 'Aaron Boone',
'1973-03-09', 'USA');

INSERT INTO Manager VALUES (2001);

INSERT INTO Person VALUES (2002, 'Carlos Mendoza',
'1980-11-27', 'Venezuela');

INSERT INTO Coach VALUES (2002);

INSERT INTO Person VALUES (2003, 'Dillon Lawson',
'1985-06-15', 'USA');

INSERT INTO Coach VALUES (2003);

INSERT INTO Person VALUES (2004, 'Matt Blake', '1985-05-14',
'USA');

INSERT INTO Coach VALUES (2004);

INSERT INTO Person VALUES (2005, 'Travis Chapman',
'1978-06-05', 'USA');

```



```
INSERT INTO Coach VALUES (2005);
```

```
INSERT INTO Person VALUES (2006, 'Luis Rojas', '1981-09-01',  
'Dominican Republic');
```

```
INSERT INTO Coach VALUES (2006);
```

```
-- Toronto Blue Jays Roster
```

```
INSERT INTO Person (PersonID, Name, DateOfBirth,  
Nationality)
```

```
VALUES (1101, 'Bo Bichette', '1998-03-05', 'USA');
```

```
INSERT INTO Person VALUES (1102, 'Vladimir Guerrero Jr.',  
'1999-03-16', 'Canada');
```

```
INSERT INTO Person VALUES (1103, 'George Springer',  
'1989-09-19', 'USA');
```

```
INSERT INTO Person VALUES (1104, 'Alejandro Kirk',  
'1998-11-06', 'Mexico');
```

```
INSERT INTO Person VALUES (1105, 'Tyler Heineman',  
'1991-06-19', 'USA');
```

```
INSERT INTO Person VALUES (1106, 'Andres Gimenez',  
'1998-09-04', 'Venezuela');
```

```
INSERT INTO Person VALUES (1107, 'Ernie Clement',  
'1996-03-22', 'USA');  
  
INSERT INTO Person VALUES (1108, 'Will Wagner',  
'1998-07-29', 'USA');  
  
INSERT INTO Person VALUES (1109, 'Anthony Santander',  
'1994-10-19', 'Venezuela');  
  
INSERT INTO Person VALUES (1110, 'Myles Straw',  
'1994-10-17', 'USA');  
  
INSERT INTO Person VALUES (1111, 'Nathan Lukes',  
'1994-07-12', 'USA');  
  
INSERT INTO Person VALUES (1112, 'Alan Roden', '1999-04-29',  
'USA');  
  
INSERT INTO Person VALUES (1113, 'Davis Schneider',  
'1999-06-03', 'USA');  
  
INSERT INTO Person VALUES (1114, 'José Berríos',  
'1994-05-27', 'Puerto Rico');  
  
INSERT INTO Person VALUES (1115, 'Kevin Gausman',  
'1991-01-06', 'USA');  
  
INSERT INTO Person VALUES (1116, 'Max Scherzer',  
'1984-07-27', 'USA');
```

```

INSERT INTO Person VALUES (1117, 'Chris Bassitt',
'1989-02-22', 'USA');

INSERT INTO Person VALUES (1118, 'Bowden Francis',
'1996-04-18', 'USA');

INSERT INTO Person VALUES (1119, 'Nick Sandlin',
'1997-01-10', 'USA');

INSERT INTO Person VALUES (1120, 'Yimi García',
'1990-08-18', 'Dominican Republic');

INSERT INTO Person VALUES (1121, 'Jeff Hoffman',
'1993-01-08', 'USA');

INSERT INTO Person VALUES (1122, 'Nate Pearson',
'1996-08-20', 'USA');

INSERT INTO Person VALUES (1123, 'Wes Parsons',
'1992-09-06', 'USA');

INSERT INTO Player VALUES (1101, 'Shortstop', 'Right',
'Right');

INSERT INTO Player VALUES (1102, 'First Base', 'Right',
'Right');

INSERT INTO Player VALUES (1103, 'Outfield', 'Right',
'Right');

```

```
INSERT INTO Player VALUES (1104, 'Catcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1105, 'Catcher', 'Switch',  
'Right');  
  
INSERT INTO Player VALUES (1106, 'Second Base', 'Left',  
'Right');  
  
INSERT INTO Player VALUES (1107, 'Utility', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1108, 'Infield', 'Left',  
'Right');  
  
INSERT INTO Player VALUES (1109, 'Outfield', 'Switch',  
'Right');  
  
INSERT INTO Player VALUES (1110, 'Outfield', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1111, 'Outfield', 'Left',  
'Left');  
  
INSERT INTO Player VALUES (1112, 'Outfield', 'Left',  
'Right');  
  
INSERT INTO Player VALUES (1113, 'Infield', 'Right',  
'Right');
```

```
INSERT INTO Player VALUES (1114, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1115, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1116, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1117, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1118, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1119, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1120, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1121, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1122, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1123, 'Pitcher', 'Right',  
'Right');
```

-- New York Yankees Roster

```
INSERT INTO Person VALUES (1201, 'Aaron Judge',  
'1992-04-26', 'USA');
```

```
INSERT INTO Person VALUES (1202, 'Giancarlo Stanton',  
'1989-11-08', 'USA');
```

```
INSERT INTO Person VALUES (1203, 'Paul Goldschmidt',  
'1987-09-10', 'USA');
```

```
INSERT INTO Person VALUES (1204, 'Jazz Chisholm Jr.',  
'1998-02-01', 'Bahamas');
```

```
INSERT INTO Person VALUES (1205, 'Anthony Volpe',  
'2001-04-28', 'USA');
```

```
INSERT INTO Person VALUES (1206, 'Oswaldo Cabrera',  
'1999-03-01', 'Venezuela');
```

```
INSERT INTO Person VALUES (1207, 'Ben Rice', '1998-02-09',  
'USA');
```

```
INSERT INTO Person VALUES (1208, 'Cody Bellinger',  
'1995-07-13', 'USA');
```

```
INSERT INTO Person VALUES (1209, 'Jasson Domínguez',  
'2003-02-07', 'Dominican Republic');
```

```
INSERT INTO Person VALUES (1210, 'Alex Verdugo',  
'1996-05-15', 'USA');  
  
INSERT INTO Person VALUES (1211, 'Trent Grisham',  
'1996-11-01', 'USA');  
  
INSERT INTO Person VALUES (1212, 'Austin Wells',  
'1999-07-12', 'USA');  
  
INSERT INTO Person VALUES (1213, 'J.C. Escarra',  
'1994-11-27', 'USA');  
  
INSERT INTO Person VALUES (1214, 'Carlos Rodón',  
'1992-12-10', 'USA');  
  
INSERT INTO Person VALUES (1215, 'Clarke Schmidt',  
'1996-02-20', 'USA');  
  
INSERT INTO Person VALUES (1216, 'Max Fried', '1994-01-18',  
'USA');  
  
INSERT INTO Person VALUES (1217, 'Carlos Carrasco',  
'1987-03-21', 'Venezuela');  
  
INSERT INTO Person VALUES (1218, 'Will Warren',  
'1999-06-16', 'USA');  
  
INSERT INTO Person VALUES (1219, 'Ian Hamilton',  
'1995-06-16', 'USA');
```

```

INSERT INTO Person VALUES (1220, 'Tim Hill', '1990-02-10',
'USA');

INSERT INTO Person VALUES (1221, 'Mark Leiter Jr.',
'1991-03-13', 'USA');

INSERT INTO Person VALUES (1222, 'Tyler Matzek',
'1990-10-19', 'USA');

INSERT INTO Person VALUES (1223, 'Devin Williams',
'1994-09-21', 'USA');

INSERT INTO Person VALUES (1224, 'Ryan Yarbrough',
'1992-12-31', 'USA');

INSERT INTO Player VALUES (1201, 'Outfield', 'Right',
'Right');

INSERT INTO Player VALUES (1202, 'Outfield', 'Right',
'Right');

INSERT INTO Player VALUES (1203, 'First Base', 'Right',
'Right');

INSERT INTO Player VALUES (1204, 'Second Base', 'Left',
'Right');

INSERT INTO Player VALUES (1205, 'Shortstop', 'Right',
'Right');

```



```
INSERT INTO Player VALUES (1206, 'Utility', 'Switch',  
'Right');  
  
INSERT INTO Player VALUES (1207, 'Catcher', 'Left',  
'Right');  
  
INSERT INTO Player VALUES (1208, 'Outfield', 'Left',  
'Left');  
  
INSERT INTO Player VALUES (1209, 'Outfield', 'Switch',  
'Right');  
  
INSERT INTO Player VALUES (1210, 'Outfield', 'Left',  
'Left');  
  
INSERT INTO Player VALUES (1211, 'Outfield', 'Left',  
'Left');  
  
INSERT INTO Player VALUES (1212, 'Catcher', 'Left',  
'Right');  
  
INSERT INTO Player VALUES (1213, 'Catcher', 'Left',  
'Right');  
  
INSERT INTO Player VALUES (1214, 'Pitcher', 'Left', 'Left');  
INSERT INTO Player VALUES (1215, 'Pitcher', 'Right',  
'Right');  
  
INSERT INTO Player VALUES (1216, 'Pitcher', 'Left', 'Left');
```

```

INSERT INTO Player VALUES (1217, 'Pitcher', 'Right',
'Right');

INSERT INTO Player VALUES (1218, 'Pitcher', 'Right',
'Right');

INSERT INTO Player VALUES (1219, 'Pitcher', 'Right',
'Right');

INSERT INTO Player VALUES (1220, 'Pitcher', 'Left', 'Left');
INSERT INTO Player VALUES (1221, 'Pitcher', 'Right',
'Right');

INSERT INTO Player VALUES (1222, 'Pitcher', 'Left', 'Left');
INSERT INTO Player VALUES (1223, 'Pitcher', 'Right',
'Right');

INSERT INTO Player VALUES (1224, 'Pitcher', 'Left', 'Left');

-- CONTRACTS

INSERT INTO Contract (ContractID, PlayerID, TeamID,
StartDate, EndDate, Salary) VALUES
(1, 1101, 1, '2023-04-01', '2026-10-31', 12500000.00);

INSERT INTO Contract VALUES (2, 1102, 1, '2022-03-01',
'2028-10-31', 14500000.00);

```

```

INSERT INTO Contract VALUES (3, 1201, 2, '2021-02-15',
'2029-10-31', 30000000.00);

INSERT INTO Contract VALUES (4, 1202, 2, '2020-01-10',
'2025-10-31', 25000000.00);

-- PLAYER STATS for GameID 1001

INSERT INTO PlayerStats (PlayerID, GameID, AtBats, Hits,
Runs, HomeRuns, RBIs, Strikeouts, Walks, Errors,
InningsPitched) VALUES
(1101, 1001, 4, 2, 1, 1, 2, 0, 1, 0, 0.0);

INSERT INTO PlayerStats VALUES (1102, 1001, 3, 1, 0, 0, 1,
1, 1, 0, 0.0);

INSERT INTO PlayerStats VALUES (1201, 1001, 4, 1, 1, 1, 1,
1, 0, 0, 0.0);

INSERT INTO PlayerStats VALUES (1202, 1001, 3, 0, 0, 0, 0,
2, 0, 0, 0.0);

```

Table and Records

DESC Person;

	Field	Type	Null	Key	Default	Extra
►	PersonID	int	NO	PRI	NULL	
	Name	var...	YES		NULL	
	DateOfBirth	date	YES		NULL	
	Nationality	var...	YES		NULL	

SELECT * FROM Person;

	PersonID	Name	DateOfBirth	Nationality
►	1001	John Schneider	1980-02-14	USA
	1002	Don Mattingly	1961-04-20	USA
	1003	Guillermo Martínez	1984-11-05	USA
	1004	Pete Walker	1969-04-08	USA
	1005	Mark Budzinski	1973-08-20	USA
	1006	Luis Rivera	1964-01-03	Puerto Rico
	1101	Bo Bichette	1998-03-05	USA
	1102	Vladimir Guerrero Jr.	1999-03-16	Canada
	1103	George Springer	1989-09-19	USA
	1104	Alejandro Kirk	1998-11-06	Mexico
	1105	Tyler Heineman	1991-06-19	USA
	1106	Andres Gimenez	1998-09-04	Venezuela
	1107	Ernie Clement	1996-03-22	USA
	1108	Will Wagner	1998-07-29	USA
	1109	Anthony Santander	1994-10-19	Venezuela
	1110	Myles Straw	1994-10-17	USA
	1111	Nathan Lukes	1994-07-12	USA
	1112	Alan Roden	1999-04-29	USA
	1113	Davis Schneider	1999-06-03	USA
	1114	José Berríos	1994-05-27	Puerto Rico
	1115	Kevin Gausman	1991-01-06	USA
	1116	Max Scherzer	1984-07-27	USA
	1117	Chris Bassitt	1989-02-22	USA
	1118	Bowden Francis	1996-04-18	USA

	1119	Nick Sandlin	1997-01-10	USA
	1120	Yimi García	1990-08-18	Dominican ...
	1121	Jeff Hoffman	1993-01-08	USA
	1122	Nate Pearson	1996-08-20	USA
	1123	Wes Parsons	1992-09-06	USA
	1201	Aaron Judge	1992-04-26	USA
	1202	Giancarlo Stanton	1989-11-08	USA
	1203	Paul Goldschmidt	1987-09-10	USA
	1204	Jazz Chisholm Jr.	1998-02-01	Bahamas
	1205	Anthony Volpe	2001-04-28	USA
	1206	Oswaldo Cabrera	1999-03-01	Venezuela
	1207	Ben Rice	1998-02-09	USA
	1208	Cody Bellinger	1995-07-13	USA
	1209	Jasson Domínguez	2003-02-07	Dominican ...
	1210	Alex Verdugo	1996-05-15	USA
	1211	Trent Grisham	1996-11-01	USA
	1212	Austin Wells	1999-07-12	USA
	1213	J.C. Escarra	1994-11-27	USA
	1214	Carlos Rodón	1992-12-10	USA
	1215	Clarke Schmidt	1996-02-20	USA
	1216	Max Fried	1994-01-18	USA
	1217	Carlos Carrasco	1987-03-21	Venezuela
	1218	Will Warren	1999-06-16	USA
	1219	Ian Hamilton	1995-06-16	USA
	1220	Tim Hill	1990-02-10	USA
	1221	Mark Leiter Jr.	1991-03-13	USA
	1222	Tyler Matzek	1990-10-19	USA
	1223	Devin Williams	1994-09-21	USA
	1224	Ryan Yarbrough	1992-12-31	USA
	2001	Aaron Boone	1973-03-09	USA
	2002	Carlos Mendoza	1980-11-27	Venezuela
	2003	Dillon Lawson	1985-06-15	USA
	2004	Matt Blake	1985-05-14	USA
	2005	Travis Chapman	1978-06-05	USA
	2006	Luis Rojas	1981-09-01	Dominican ...
•	NULL	NULL	NULL	NULL

DESC Player;

	Field	Type	Null	Key	Default	Extra
►	PersonID	int	NO	PRI	NULL	
	Position	var...	YES		NULL	
	Bats	var...	YES		NULL	
	Throws	var...	YES		NULL	

SELECT * FROM Player;

	PersonID	Position	Bats	Throws
►	1101	Shortstop	Right	Right
	1102	First Base	Right	Right
	1103	Outfield	Right	Right
	1104	Catcher	Right	Right
	1105	Catcher	Switch	Right
	1106	Second Base	Left	Right
	1107	Utility	Right	Right
	1108	Infield	Left	Right
	1109	Outfield	Switch	Right
	1110	Outfield	Right	Right
	1111	Outfield	Left	Left
	1112	Outfield	Left	Right
	1113	Infield	Right	Right
	1114	Pitcher	Right	Right
	1115	Pitcher	Right	Right
	1116	Pitcher	Right	Right
	1117	Pitcher	Right	Right
	1118	Pitcher	Right	Right
	1119	Pitcher	Right	Right
	1120	Pitcher	Right	Right
	1121	Pitcher	Right	Right
	1122	Pitcher	Right	Right
	1123	Pitcher	Right	Right
	1201	Outfield	Right	Right
	1202	Outfield	Right	Right
	1203	First Base	Right	Right
	1204	Second Base	Left	Right
	1205	Shortstop	Right	Right
	1206	Utility	Switch	Right
	1207	Catcher	Left	Right
	1208	Outfield	Left	Left
	1209	Outfield	Switch	Right
	1210	Outfield	Left	Left
	1211	Outfield	Left	Left
	1212	Catcher	Left	Right
	1213	Catcher	Left	Right
	1214	Pitcher	Left	Left
	1215	Pitcher	Right	Right
	1216	Pitcher	Left	Left
	1217	Pitcher	Right	Right
	1218	Pitcher	Right	Right
	1219	Pitcher	Right	Right
	1220	Pitcher	Left	Left
	1221	Pitcher	Right	Right
	1222	Pitcher	Left	Left
	1223	Pitcher	Right	Right
	1224	Pitcher	Left	Left
	NULL	NULL	NULL	NULL

DESC Coach;

	Field	Type	Null	Key	Default	Extra
►	PersonID	int	NO	PRI	NULL	

SELECT * FROM Coach;

	PersonID
►	1002
	1003
	1004
	1005
	1006
	2002
	2003
	2004
	2005
	2006
✱	NULL

DESC Manager;

	Field	Type	Null	Key	Default	Extra
►	PersonID	int	NO	PRI	NULL	

SELECT * FROM Manager;

	PersonID
►	1001
	2001
✱	NULL

DESC Team;

	Field	Type	Null	Key	Default	Extra
▶	TeamID	int	NO	PRI	NULL	
	TeamName	var...	YES		NULL	
	City	var...	YES		NULL	
	Division	var...	YES		NULL	
	League	var...	YES		NULL	

SELECT * FROM Team;

	TeamID	TeamName	City	Division	League
▶	1	Toronto Blue Jays	Toronto	AL East	American
	2	New York Yankees	New York	AL East	American
*	NULL	NULL	NULL	NULL	NULL

DESC Contract;

	Field	Type	Null	Key	Default	Extra
▶	ContractID	int	NO	PRI	NULL	
	PlayerID	int	YES	MUL	NULL	
	TeamID	int	YES	MUL	NULL	
	StartDate	date	YES		NULL	
	EndDate	date	YES		NULL	
	Salary	de...	YES		NULL	

SELECT * FROM Contract;

	ContractID	PlayerID	TeamID	StartDate	EndDate	Salary
▶	1	1101	1	2023-04-01	2026-10-31	12500000.00
	2	1102	1	2022-03-01	2028-10-31	14500000.00
	3	1201	2	2021-02-15	2029-10-31	30000000.00
	4	1202	2	2020-01-10	2025-10-31	25000000.00
*	NULL	NULL	NULL	NULL	NULL	NULL

DESC Season;

	Field	Type	Null	Key	Default	Extra
▶	SeasonID	int	NO	PRI	NULL	
	Year	int	YES		NULL	
	LeagueChampion	var...	YES		NULL	
	MVP	var...	YES		NULL	

SELECT * FROM Season;

	SeasonID	Year	LeagueChampion	MVP
▶	2023	2023	Texas Rangers	Shohei Ohtani
	2024	2024	Los Angeles Dodgers	Shohei Ohtani
*	NULL	NULL	NULL	NULL

DESC Game;

	Field	Type	Null	Key	Default	Extra
▶	GameID	int	NO	PRI	NULL	
	Date	date	YES		NULL	
	HomeTeamID	int	YES	MUL	NULL	
	AwayTeamID	int	YES	MUL	NULL	
	Stadium	var...	YES		NULL	
	FinalScore	var...	YES		NULL	
	SeasonID	int	YES	MUL	NULL	

SELECT * FROM Game;

	GameID	Date	HomeTeamID	AwayTeamID	Stadium	FinalScore	SeasonID
▶	1001	2024-03-28	1	2	Rogers Centre	5-3	2024
	1002	2024-03-30	2	1	Yankee Stadium	4-2	2024
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL


```
DESC PlayerStats;
```

	Field	Type	Null	Key	Default	Extra
►	PlayerID	int	NO	PRI	NULL	
	GameID	int	NO	PRI	NULL	
	AtBats	int	YES		NULL	
	Hits	int	YES		NULL	
	Runs	int	YES		NULL	
	HomeRuns	int	YES		NULL	
	RBIs	int	YES		NULL	
	Strikeouts	int	YES		NULL	
	Walks	int	YES		NULL	
	Errors	int	YES		NULL	
	InningsPit...	de...	YES		NULL	

```
SELECT * FROM PlayerStats;
```

Queries

1. List all players in the database

```
SELECT Name FROM Person
WHERE PersonID IN (SELECT PersonID FROM Player);
```

Name	
Bo Bichette	Giancarlo Stanton
Vladimir Guerrero Jr.	Paul Goldschmidt
George Springer	Jazz Chisholm Jr.
Alejandro Kirk	Anthony Volpe
Tyler Heineman	Oswaldo Cabrera
Andres Gimenez	Ben Rice
Ernie Clement	Cody Bellinger
Will Wagner	Jasson Domínguez
Anthony Santander	Alex Verdugo
Myles Straw	Trent Grisham
Nathan Lukes	Austin Wells
Alan Roden	J.C. Escarra
Davis Schneider	Carlos Rodón
José Berríos	Clarke Schmidt
Kevin Gausman	Max Fried
Max Scherzer	Carlos Carrasco
Chris Bassitt	Will Warren
Bowden Francis	Ian Hamilton
Nick Sandlin	Tim Hill
Yimi García	Mark Leiter Jr.
Jeff Hoffman	Tyler Matzek
Nate Pearson	Devin Williams
Wes Parsons	Ryan Yarbrough
Aaron Judge	

2. Find Bo Bichette's contract details

```
SELECT p.Name, c.StartDate, c.EndDate, c.Salary
FROM Person p
JOIN Contract c ON p.PersonID = c.PlayerID
WHERE p.Name = 'Bo Bichette';
```

Name	StartDate	EndDate	Salary
Bo Bichette	2023-04-01	2026-10-31	12500000.00

3. Show players without any stats recorded yet

```
SELECT p.Name
FROM Player pl
JOIN Person p ON pl.PersonID = p.PersonID
LEFT JOIN PlayerStats ps ON pl.PersonID = ps.PlayerID
WHERE ps.PlayerID IS NULL;
```

	Name
▶	George Springer
	Alejandro Kirk
	Tyler Heineman
	Andres Gimenez
	Ernie Clement
	Will Wagner
	Anthony Santander
	Myles Straw
	Nathan Lukes
	Alan Roden
	Davis Schneider
	José Berríos
	Kevin Gausman
	Max Scherzer
	Chris Bassitt
	Bowden Francis
	Nick Sandlin
	Yimi García
	Jeff Hoffman
	Nate Pearson
	Wes Parsons

	Paul Goldschmidt
	Jazz Chisholm Jr.
	Anthony Volpe
	Oswaldo Cabrera
	Ben Rice
	Cody Bellinger
	Jasson Domínguez
	Alex Verdugo
	Trent Grisham
	Austin Wells
	J.C. Escarra
	Carlos Rodón
	Clarke Schmidt
	Max Fried
	Carlos Carrasco
	Will Warren
	Ian Hamilton
	Tim Hill
	Mark Leiter Jr.
	Tyler Matzek
	Devin Williams
	Ryan Yarbrough

4. Find all players who played in Game 1001

```
SELECT p.Name
FROM PlayerStats ps
JOIN Player pl ON ps.PlayerID = pl.PersonID
JOIN Person p ON p.PersonID = pl.PersonID
WHERE ps.GameID = 1001;
```

	Name
▶	Bo Bichette
	Vladimir Guerrero Jr.
	Aaron Judge
	Giancarlo Stanton

5. List games played between Blue Jays and Yankees

```
SELECT GameID, Date, FinalScore
FROM Game
WHERE (HomeTeamID = 1 AND AwayTeamID = 2) OR (HomeTeamID = 2
AND AwayTeamID = 1);
```

	GameID	Date	FinalScore
▶	1001	2024-03-28	5-3
	1002	2024-03-30	4-2
*	NULL	NULL	NULL

6. Find players who hit at least one home run in Game 1001

```
SELECT p.Name, ps.HomeRuns
FROM PlayerStats ps
JOIN Player pl ON ps.PlayerID = pl.PersonID
JOIN Person p ON p.PersonID = pl.PersonID
```

```
WHERE ps.GameID = 1001 AND ps.HomeRuns > 0;
```

	Name	HomeRuns
▶	Bo Bichette	1
	Aaron Judge	1

7. Show the average salary per team

```
SELECT t.TeamName, AVG(c.Salary) AS AvgSalary
FROM Contract c
JOIN Team t ON c.TeamID = t.TeamID
GROUP BY t.TeamName;
```

	TeamName	AvgSalary
▶	Toronto Blue Jays	13500000.000000
	New York Yankees	27500000.000000

8. Count number of players per team (with contracts)

```
SELECT t.TeamName, COUNT(c.PlayerID) AS PlayerCount
FROM Contract c
JOIN Team t ON c.TeamID = t.TeamID
GROUP BY t.TeamName;
```

	TeamName	PlayerCount
▶	Toronto Blue Jays	2
	New York Yankees	2

9. Show players and total RBIs (across all games)

```
SELECT p.Name, SUM(ps.RBIs) AS TotalRBIs
FROM PlayerStats ps
JOIN Person p ON ps.PlayerID = p.PersonID
GROUP BY p.Name
ORDER BY TotalRBIs DESC;
```

	Name	TotalRBIs
▶	Bo Bichette	2
	Vladimir Guerrero Jr.	1
	Aaron Judge	1
	Giancarlo Stanton	0

10. List all games with team names

	GameID	Date	HomeTeam	AwayTeam	FinalScore
▶	1002	2024-03-30	New York Yankees	Toronto Blue Jays	4-2
	1001	2024-03-28	Toronto Blue Jays	New York Yankees	5-3